The 1D advection–diffusion equation

Ismael Martín Díez
Engineering Fluid Mechanics — Aerospace Engineering, Universidad Carlos III de Madrid
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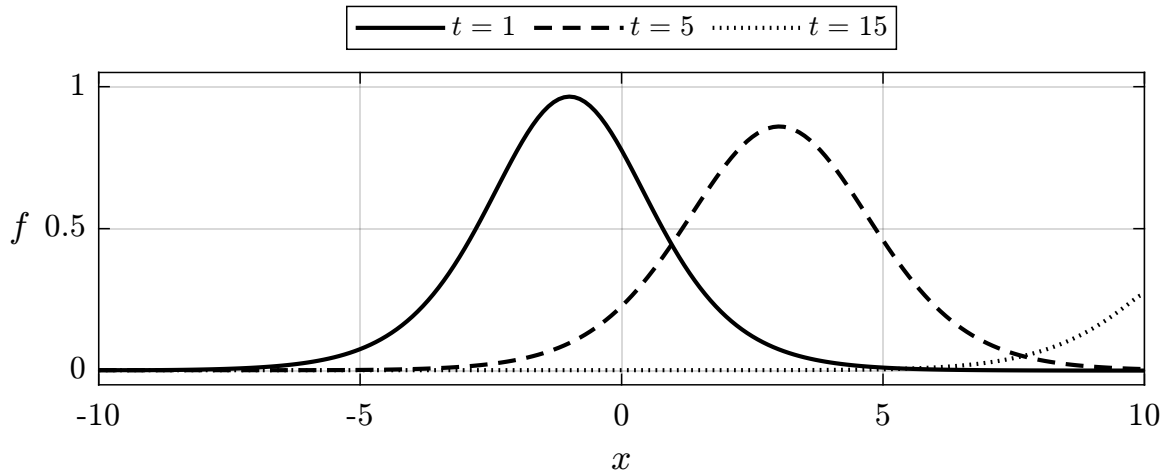


Figure 1: $f(x,t)$ with Neumann boundary conditions ($\partial_x f = 0$ at $x = \pm 10$) at $t = 1$, 5 and 15. Parameters: $U = 1$, $\alpha = 0.1$ ($k = 0.5$). The pulse starts centered at $x = -2$ and the flow pushes it to the right at speed U. Its peak is at $x \simeq -1$ for $t = 1$ and at $x \simeq 3$ for $t = 5$. On the way it gets a bit lower because diffusion spreads it out. At $t = 15$ it has reached the right end so the highest value is now right at the wall and most of the pulse is already gone.

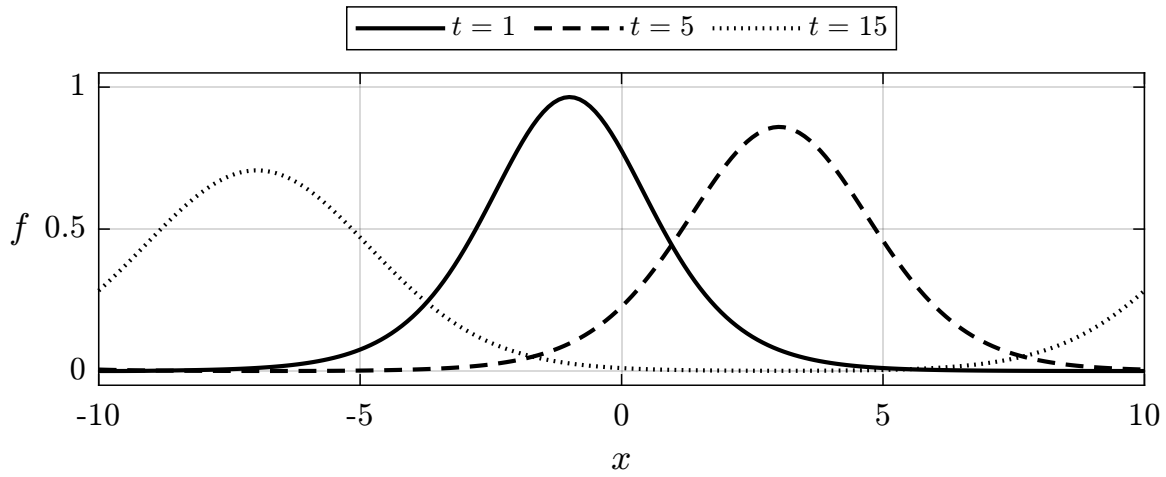


Figure 2: $f(x,t)$ with periodic boundary conditions ($x = -10$ and $x = 10$ are the same point) at $t = 1$, 5 and 15. Parameters: $U = 1$, $\alpha = 0.1$ ($k = 0.5$). At $t = 1$ and $t = 5$ the curves look almost the same as in Fig. 1 because the pulse has not hit either boundary yet. The difference can be seen at $t = 15$ when the pulse should be at $x = 13$ (which is outside the domain), so it re-enters from the left and its peak appears at $x \simeq -7$. In this case nothing leaves through the boundaries so the only thing diffusion does is make the pulse lower and wider.